A guilty gait: Human judgments of deception versus a computer algorithm

Helen Devine and Kurt Gray

THE UNIVERSITY of NORTH CAROLINA at CHAPEL HILL

Deepest Beliefs Lab

BACKGROUND

→ Gait can be used to determine guilt from a distance

- Jurors trust gait (or walking style) as a measure of perpetrator identification[1]
- Gait can be analyzed from a distance with CCTV footage, which is increasing in prevalence[2]

→ Human gait analysis in the courtroom is flawed

- Human judgments of guilt and identification are often fallible and biased despite our confidence[3,4]
- Forensic gait analysis in the courtroom has been subject to debate[5]
- Research on the reliability of human expert testimony on gait identification is limited[6]

→ AI algorithms judging guilt from gait are highly accurate[9,10]

- Researchers have begun to use artificial intelligence (AI) to detect deception from CCTV footage in airports[7] and retail stores[8]

OBJECTIVE

How do human judgments of guilt from gait compare to a computer algorithm's accuracy?

CONTACT:

devineh@unc.edu

@helenldevine

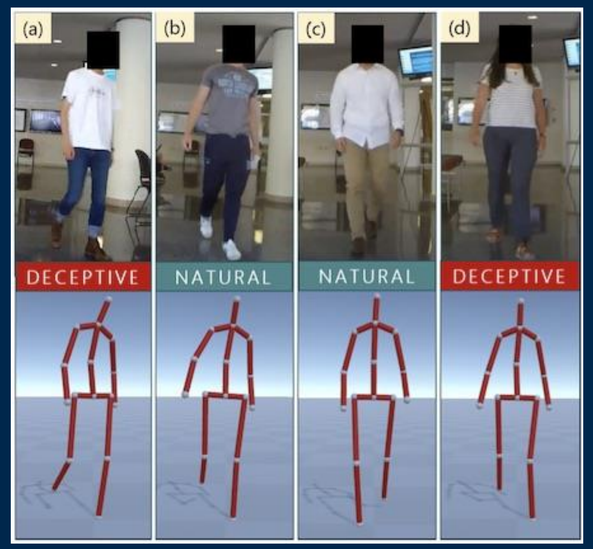
helendevine.owlstown.net

1. Stornell et al., 2021. 2. Bischoff, 2024. 3. Loftus, 2010. 4. Reinhard et al., 2013. 5. Nirenberg et al., 2018. 6. Macoveciuc et al., 2019. 7. Sujkowski et al., 2023. 8. Ansari & Singh, 2022. 9. Alghowinem et al., 2024 10. Randhavane et al., 2020).

ORIGINAL STUDY (Randhavane et al., 2020)

- N = 146 (M_{age} = 20.4, 109 female, 49 male)
- Participants were video recorded walking in one of 2 conditions:
  - a ***neutral*** condition where they carried a neutral package ("computer hardware") and were told not to drop it OR
  - a ***deceptive*** condition where they carried a large stack of cash and a mysterious package and were told to ensure no one noticed them
- An algorithm was trained to determine deception using 90% of the videos and tested on the remaining 10%.

Computer/AI Judgments ↑



Human Judgments ↓

PREREGISTERED

PRESENT STUDY

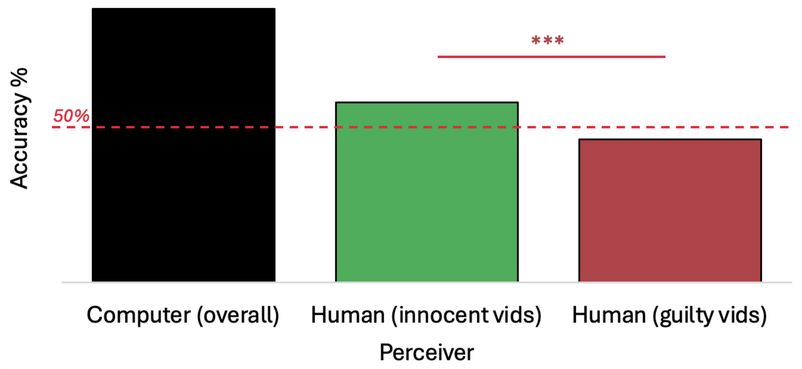
- N = 350 (M_{age} = 39.8, SD_{age} = 12.7)
  - 54.6% male, 43.1% female, 1.4% nonbinary
  - 68.5% white, 10.5% Asian, 9% Black, 9% Hispanic
- Participants watched 7 random videos from the original study. They assessed whether the subject was guilty or innocent and how certain they were of their judgment.

RESULTS

- Determining guilt: Ps performed ***better than chance*** when judging videos where the target is ***innocent*** (57.8% correct; t = 5.41, p < .001) and ***worse than chance*** when judging videos where the target is ***guilty*** (46.3% correct; t = -2.66, p = 0.008; Figure 1).
  - The computer algorithm achieved 88.4% accuracy: much higher than human judgments generally.
- Ps were more likely to judge guilt correctly when watching innocent suspects (t = 5.75, p < .001).
- Ps were *not* more confident of their judgment between innocent and guilty suspects (t = 1.89, p = 0.06).

Figure 1.

Humans perform better than chance when judging guilt from gait only when the suspect is *innocent*



DISCUSSION

- Results suggest that intuitive judgments of guilt based on gait are biased based on suspect guilt.
- People are more accurate when detecting innocence than when detecting guilt.
- Computer algorithms developed for determining guilt from gait outperform humans consistently.
- More work is needed to see how receiving information from an algorithm changes human judgments.